

Architecture Document — ET AI News Navigator

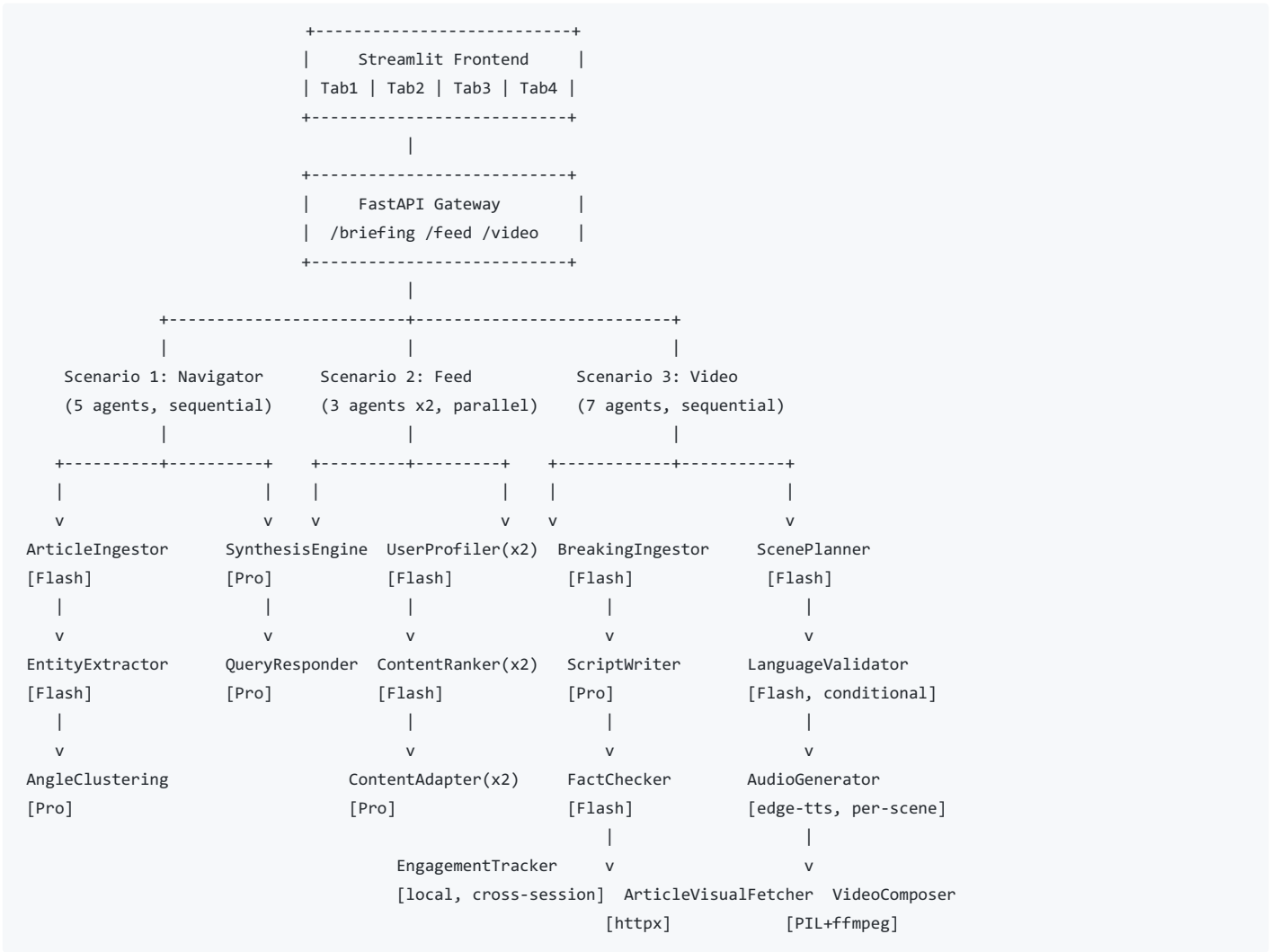
Track 8: AI-Native News Experience | ET AI Hackathon 2026

System Overview

The ET AI News Navigator is a **15-agent multi-agent system** that transforms how business news is consumed. It processes raw news articles through autonomous agent pipelines to deliver three distinct experiences:

- 1. **News Navigator** — Synthesises 22+ articles into an interactive, angle-based intelligence briefing with non-overlapping Q&A
- 2. **Personalised Feed** — Generates meaningfully different content surfaces for different user personas, with engagement-aware retuning
- 3. **Vernacular Video** — Converts breaking news into a chaptered, fact-checked explainer video in **7 Indian languages** in under 60 seconds

Architecture Diagram



Smart Model Routing (Cost Efficiency)

A deterministic routing table assigns each task type to the optimal model — **no LLM-based router overhead**. The system supports **three LLM providers** with automatic fallback:

Provider Priority: NVIDIA (primary) > Gemini (secondary) > Ollama (local fallback)

Task Type	Role	NVIDIA (Primary)	Gemini (Fallback)	Cost
Extraction, NER, classification	Flash	Llama-4 Maverick 17B	Gemini 2.0 Flash	Free / \$0.10/M

Task Type	Role	Model	Endpoint	Cost
Fact-checking, scene planning	Flash	NVIDIA (Maverick 17B)	Gemini 2.0 Flash	\$0.10/M
Language validation (conditional)	Flash	Llama-4 Maverick 17B	Gemini 2.0 Flash	Free / \$0.10/M
Multi-article synthesis	Pro	Mistral Nemotron	Gemini 2.0 Pro	Free / \$1.25/M
Creative writing (multilingual)	Pro	Mistral Nemotron	Gemini 2.0 Pro	Free / \$1.25/M
Local fallback / degradation	Ollama	qwen2.5vl:3b	—	\$0.00

With NVIDIA free endpoint: \$0.00 per pipeline run **With Gemini paid:** ~\$0.08 per run (77% savings via smart routing vs all-Pro)

Each agent step logs estimated token count and USD cost in the audit trail, making the cost saving verifiable.

Agent Roles & Communication

Scenario 1: News Navigator (5 agents + engagement tracker)

#	Agent	Model	Input	Output	Key Behaviour
1	ArticleIngestor	Flash	22 raw articles	Structured metadata + summaries	Extracts entities, categories, sentiment per article
2	EntityExtractor	Flash	Article metadata	Entity-to-article index	Normalises names, builds cross-article lookup
3	AngleClustering	Pro	Metadata + entities	5-7 angle clusters	Editorial judgment on angle separation, minimises overlap
4	SynthesisEngine	Pro	Angle clusters + full articles	Dense synthesis per angle	Non-overlapping, source-cited, with key takeaways
5	QueryResponder	Pro	Question + full history	Targeted answer	Receives all previous answers, guarantees no repetition
6	ContradictionDetector	Pro	All syntheses + articles	Cross-article contradictions	Flags opposing claims between sources with severity
7	BlindSpotDetector	Pro	Articles + angles + topic	Coverage gaps	Surfaces important topics no article covers
8	ReadingPathOptimizer	Pro	Articles + user profile	3-4 article sequence	Optimal 2-minute reading path, progressively builds understanding
9	EngagementTracker	Local	User clicks, queries, dwell time	Retuned angle ordering	Cross-session learning, no LLM cost

Inter-agent communication: Sequential pipeline with shared state. QueryResponder receives the complete synthesis history and all previous Q&A to guarantee non-overlapping answers. EngagementTracker logs interaction signals and feeds back into angle/content ranking for subsequent sessions.

Scenario 2: Personalised Feed (3 agent types x 2 personas)

#	Agent	Model	Key Behaviour
1	UserProfiler	Flash	Converts profile → content depth, format, tone, jargon level, priority topics
2	ContentRanker	Flash	Ranks articles by persona-specific relevance (0-1 score + reason)
3	ContentAdapter	Pro	Rewrites content: CFO gets exec summaries + data tables; beginner gets ELI5 + analogies

Runs in parallel for both personas. Delta summary quantifies differences (story overlap, format changes, reading level shift).

Scenario 3: Vernacular Video (5 agents, <60s budget)

#	Agent	Model	Time Budget	Key Behaviour
1	BreakingIngestor	Flash	~2s	5W1H extraction
2	ScriptWriter	Pro	~8s	Hindi script, no English jargon, cultural analogies
3	FactChecker	Flash	~3s	Claim-by-claim verification against source
4	AudioGenerator	edge-tts	~15s	Hindi TTS (hi-IN-SwaraNeural)
5	VideoComposer	PIL+ffmpeg	~15s	Visual frames + audio → MP4

Error Handling & Enterprise Readiness

Three-Tier Fallback Chain

Gemini Pro/Flash → (rate limit/error) → Retry once → Ollama qwen2.5vl:3b (local) → (if Ollama down) → Heuristic/deterministic fallback

Graceful Degradation by Scenario

Component Failure	Degraded Output	User Experience
Gemini API down	Ollama handles extraction; heuristic fallbacks for synthesis	Warning banner, partial results
LLM returns sparse data	Heuristic enrichment fills key_numbers, impact_points from article text	Complete output, slightly less nuanced
TTS voice unavailable	Fallback voice chain per language; text-only if all fail	Script readable, degraded audio
ffmpeg fails	Audio + frames returned separately	Audio playable, frames viewable
Language leakage	LanguageValidator corrects; deterministic cleanup as last resort	Clean output in target language
Timeout (>90s)	Return whatever completed	Partial results with status

Audit Trail with Cost Tracking

Every agent step logs:

- Agent name, action, model used
- Input/output summaries (truncated to 200 chars)
- Latency in milliseconds
- **Estimated token count (input + output)**
- **Estimated cost in USD**
- Status (success/fallback/error)
- Error details if applicable

Audit trails are returned with every API response and displayed in the UI with cumulative cost per pipeline run.

Technology Stack

Layer	Technology	Why
Agent Framework	Google ADK 1.18.0	SequentialAgent/ParallelAgent orchestration patterns
LLMs (Primary)	NVIDIA Mistral Nemotron + Llama-4 Maverick	Free endpoint, strong reasoning + fast extraction

LLMs (Fallback Layer)	Gemini 2.0 Pro + Flash Technology	Secondary fallback, pay-as-you-go Why
Local Fallback	Ollama + qwen2.5vl:3b	Zero-cost degradation, enterprise resilience
Backend	FastAPI 0.115.9	Async-native, OpenAPI docs
Frontend	Streamlit 1.48.1	Rapid prototyping, 4-tab interface + settings page
Graph Visualization	Plotly 5.0.0	Interactive, hoverable entity-angle network graphs
TTS	edge-tts 7.2.8	8 Indian languages with language-specific voices
Video	Pillow + ffmpeg 8.0	Chapter frames + A/V sync composition
Vector Store	ChromaDB 1.0.11	Semantic article search for Q&A
Engagement	Local JSON store	Cross-session user preference learning

Frontend UI Features

The Streamlit interface exposes all pipelines through a 4-tab design with integrated visualization and controls:

Tab 1: News Navigator

- **Briefing Generation:** Topic input with optional topic coverage enforcement
- **Angle-based Navigation:** Non-overlapping summary per angle with metadata
- **Entity Graph:** Interactive Plotly network visualization showing:
 - Entity nodes (left arc) with hoverable metadata: name, type, article count, related angles
 - Angle nodes (right lane) with connection details
 - Automatic fallback to Graphviz if Plotly unavailable
- **Entity Explorer:** Dropdown selector to drill into specific entities and their connected angles
- **Follow-up Q&A:** Ask clarifying questions on the briefing (history-aware, non-repeating answers)
- **Engagement Metrics:** Article count, angle count, pipeline latency, estimated cost

Tab 2: My ET – Create Your Profile

- **Onboarding Mode Selection:** Quick start or deep setup
- **Guided Profile Creation:** Structured onboarding questions mapped to a saved user profile
- **Profile Persistence:** Saved users become available in Tab 3 feed generation

Tab 3: Personalised Feed

- **User-Scoped Feed Generation:** Full pipeline from profile-aware subset to adapted cards
- **Sectioned Feed UX:** Headline picks, thematic sections, in-feed search
- **Article Metadata Layer:** Urgency, sentiment, credibility, freshness indicators
- **A/B Comparison:** Personalized vs baseline delta metrics and persisted Phase 7 run logging

Tab 4: Vernacular Video

- **Language Selection:** 8 languages with native voices (Hindi, Marathi, Tamil, Telugu, Kannada, Bhojpuri, Punjabi, English)
- **Video Generation:** Chapter explainer video with fact-checked narration
- **Scene Viewer:** Timeline visualization with chapter metadata

Settings Page

- **Runtime Flags:** Visibility into Retrieval Contracts and Corpus Kill Switch status
- **Ops Controls:** Crawl refresh, subset refresh, freshness metrics, recent run summaries
- **Compliance Controls:** Load compliance snapshots, generate compliance reports
- **Measurement + A/B:** View run summary, win rate, relevance lift, and recent comparison runs

Data Flow: News Navigator

[22 Raw Articles] → ArticleIngestor [Flash] → [Structured Metadata]
→ EntityExtractor [Flash] → [Entity-Article Index]
→ AngleClustering [Pro] → [5-7 Non-overlapping Angles]
→ SynthesisEngine [Pro] → [Dense Synthesis per Angle]
→ QueryResponder [Pro] ← [User Questions + Full History]
→ [Non-overlapping, Source-cited Answers]
→ EngagementTracker ← [Click/Query/Dwell Signals]

Data Flow: Vernacular Video

[Breaking Article] → BreakingIngestor [Flash] → [5W1H Facts + Key Numbers]
→ ScriptWriter [Pro] → [Language-aware Script]
→ FactChecker [Flash] → [Claim-by-claim Verification]
→ ScenePlanner [Flash] → [6-10 Chaptered Scenes + Story Arc]
→ LanguageValidator [Flash, conditional] → [Clean Target-language Output]
→ AudioGenerator [edge-tts] → [Per-scene Audio + Merged Narration]
→ VideoComposer [PIL+ffmpeg] → [Chaptered MP4 with A/V Sync]